K – Nearest Neighbors

What is Classification?

A flower shop wants to guess a customer's purchase from similarity to most recent purchase.

What is Classification?

Which flower is a customer most likely to purchase based on similarity to previous purchase?

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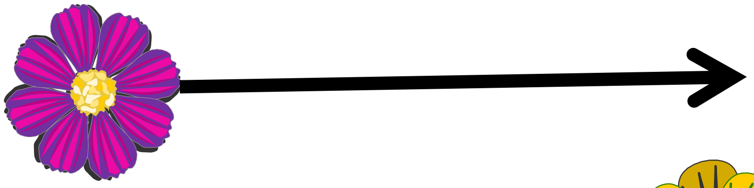

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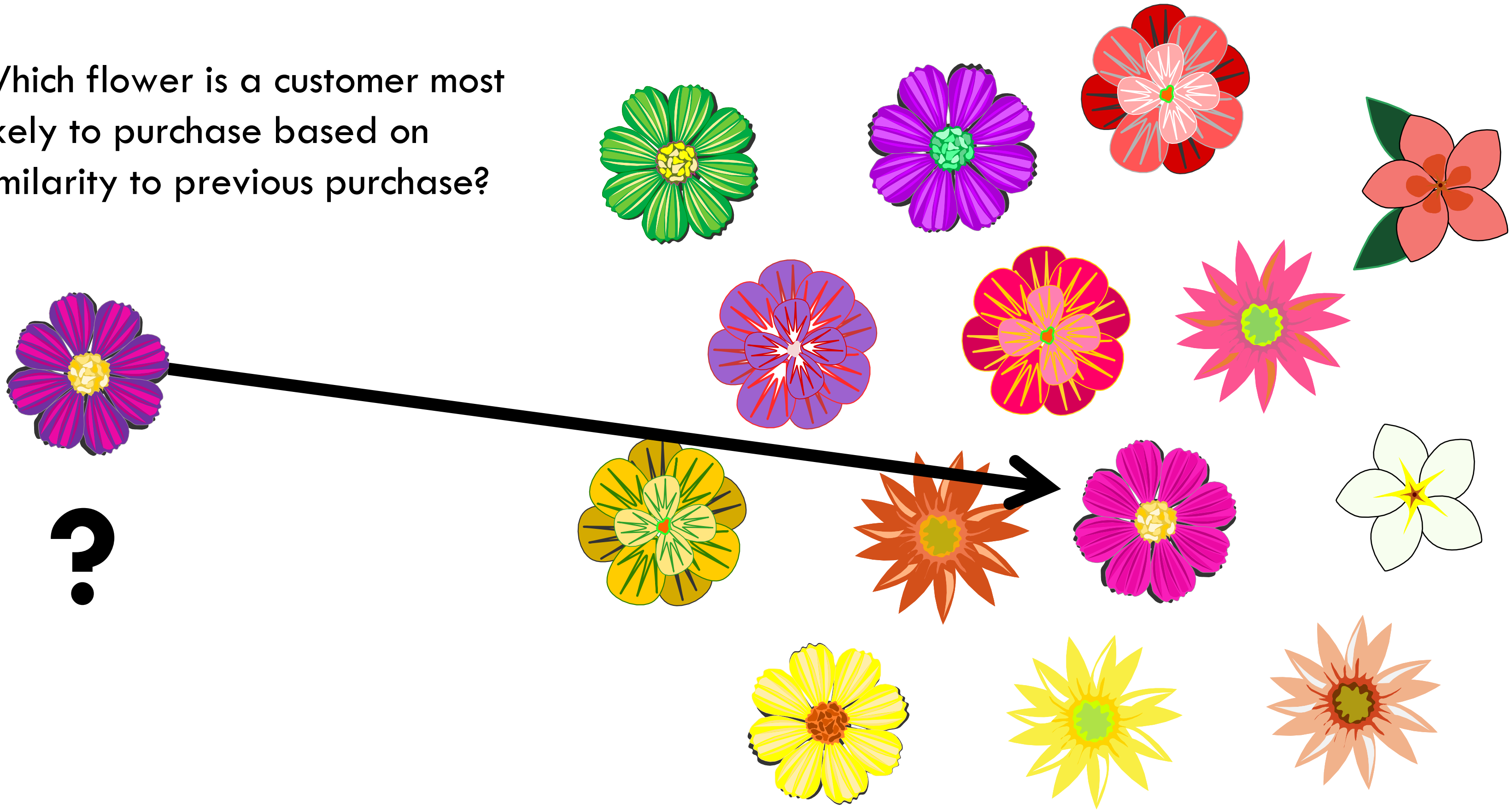

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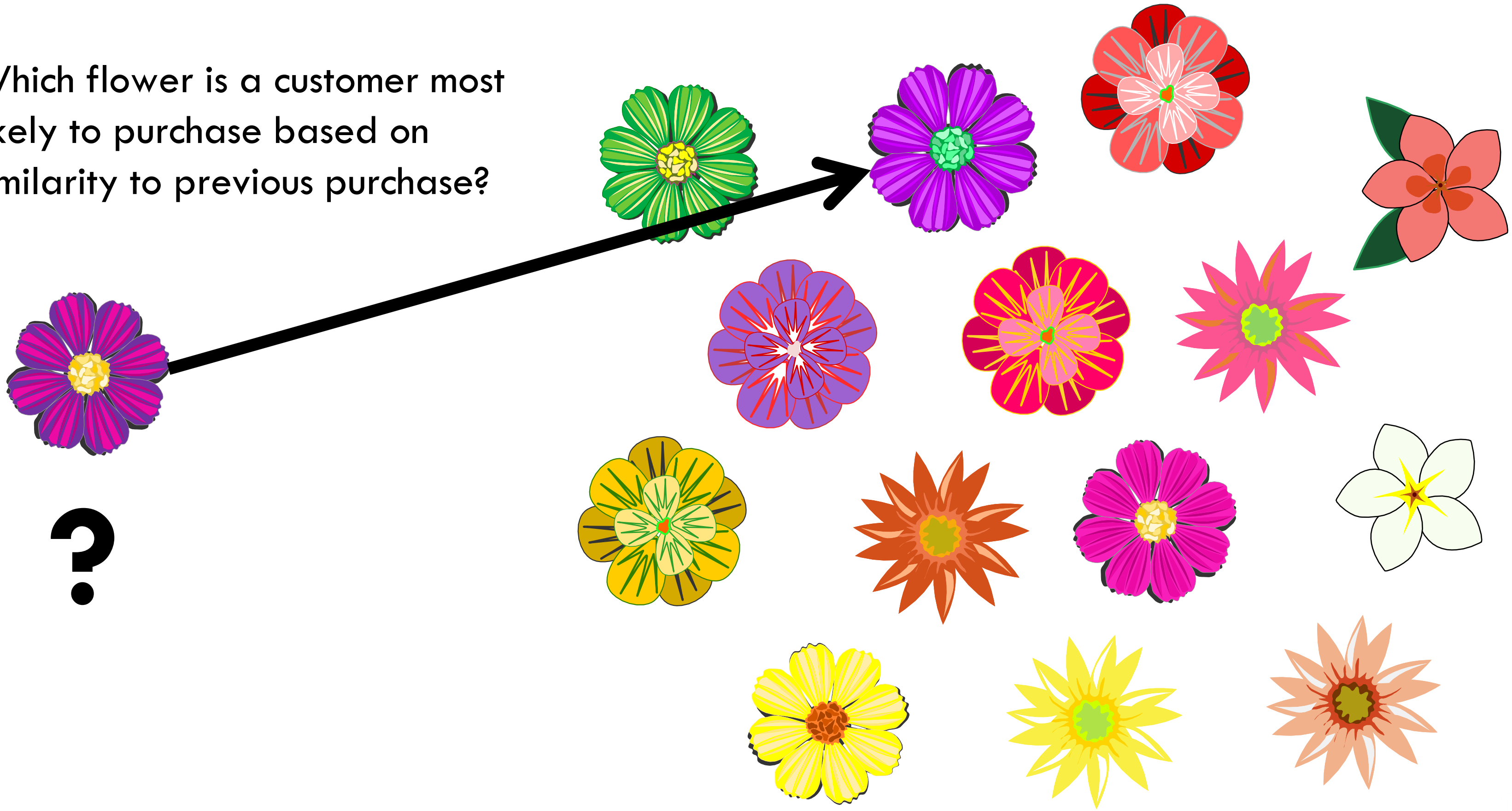

What is Needed for Classification?

- Model data with:
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- Model data with:
  - Features that can be quantitated
  - Labels that are known
- Method to measure similarity

K Nearest Neighbors Classification

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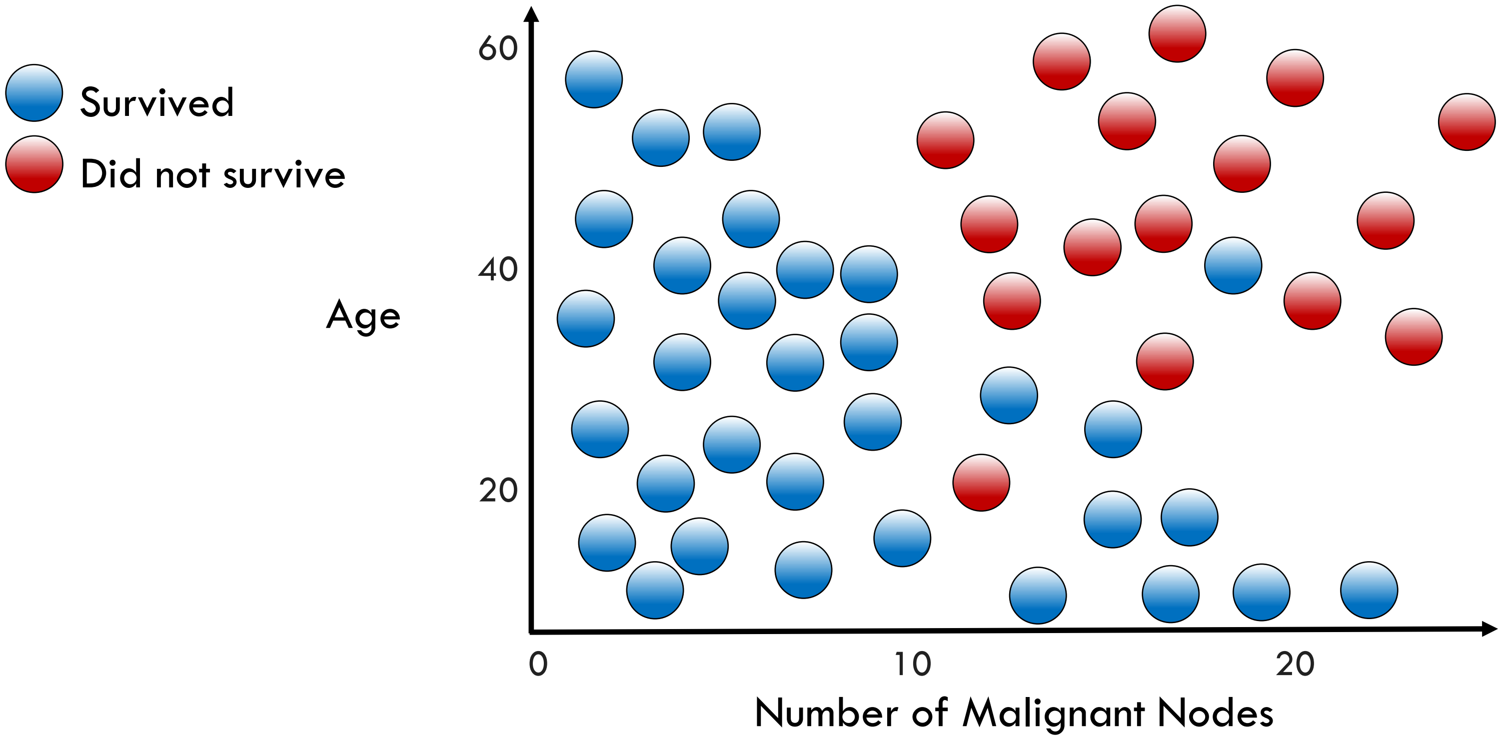

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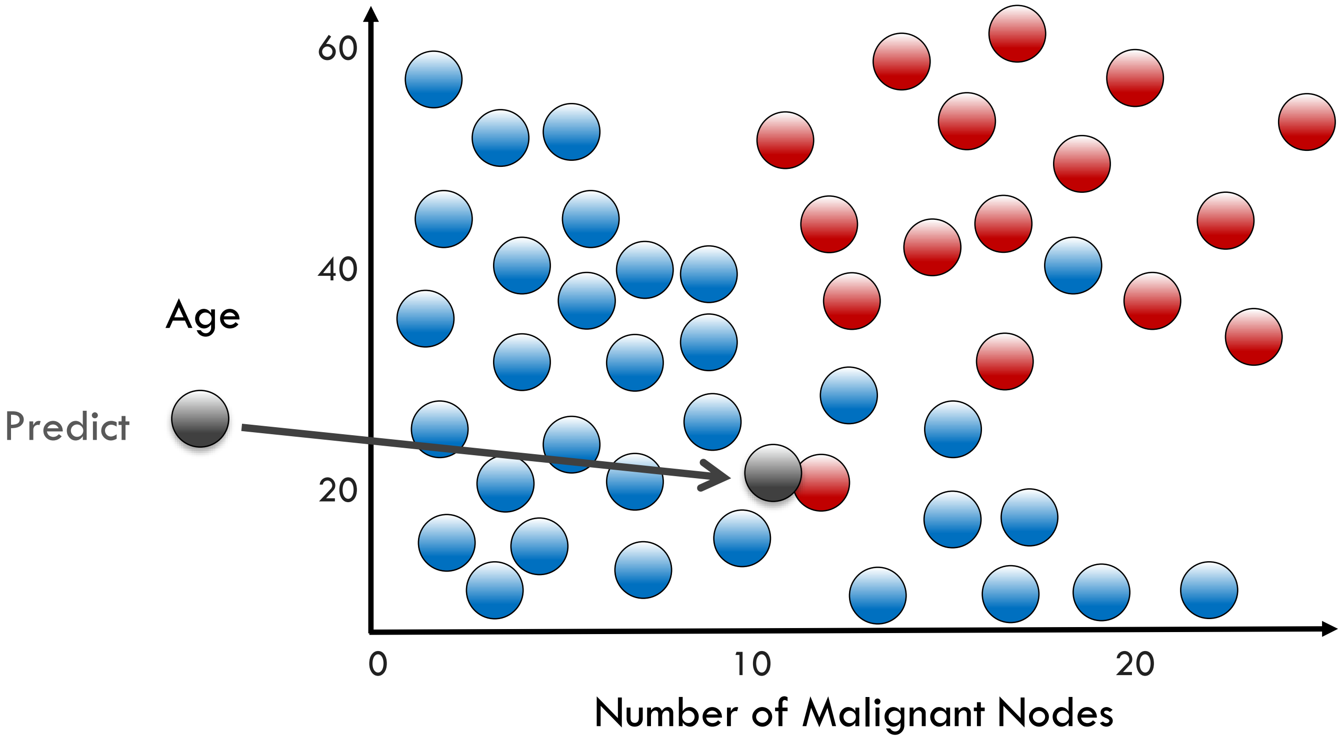
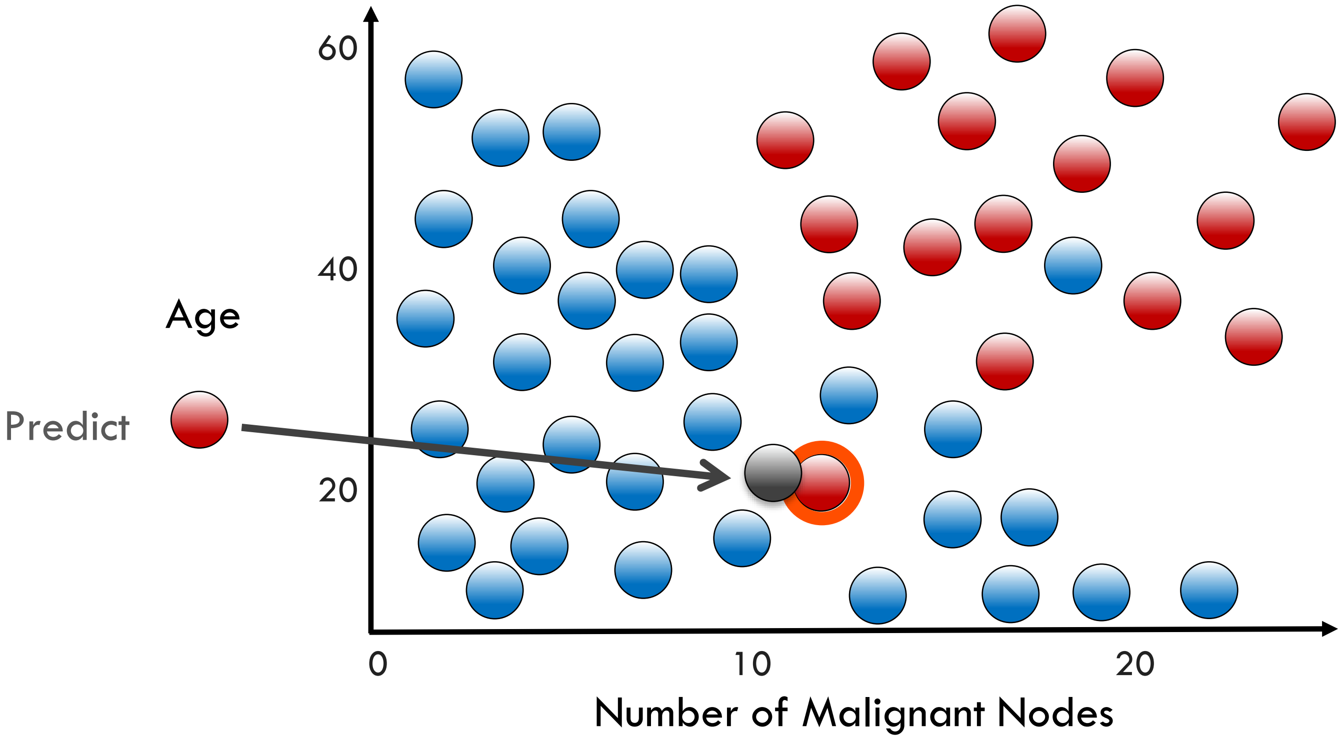


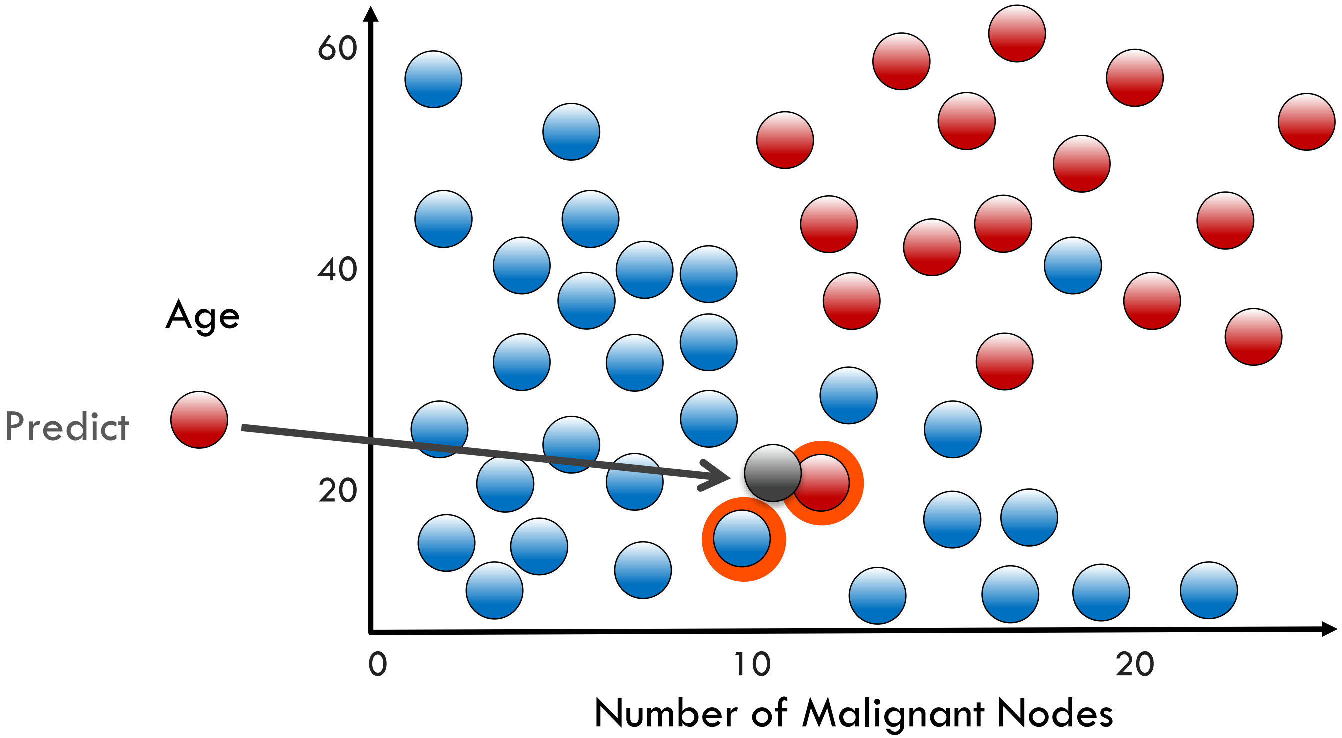
K Nearest Neighbors Classification

Neighbor Count (K = 1): 0 1

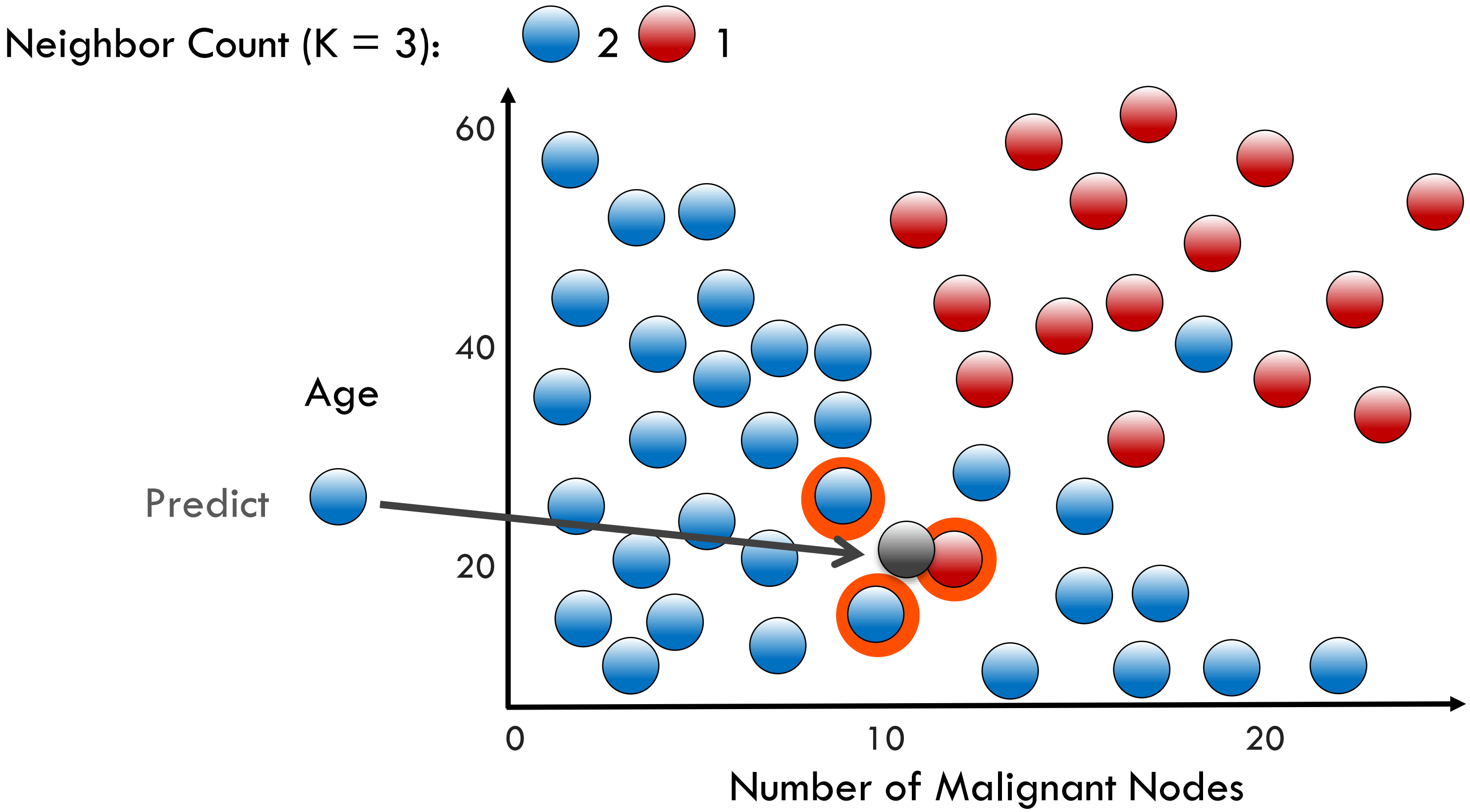


K Nearest Neighbors Classification

Neighbor Count (K = 2): 1 1

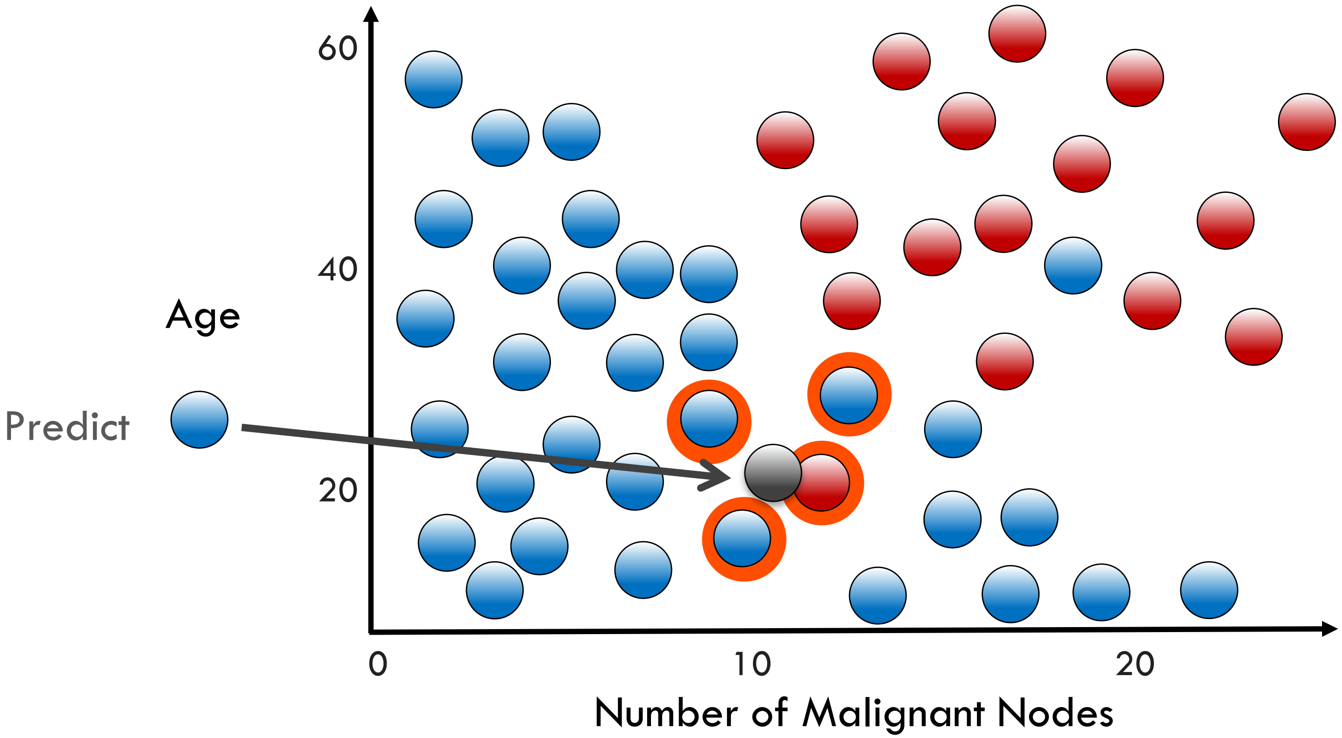


K Nearest Neighbors Classification



K Nearest Neighbors Classification

Neighbor Count (K = 4): 3 1



What is Needed to Select a KNN Model?

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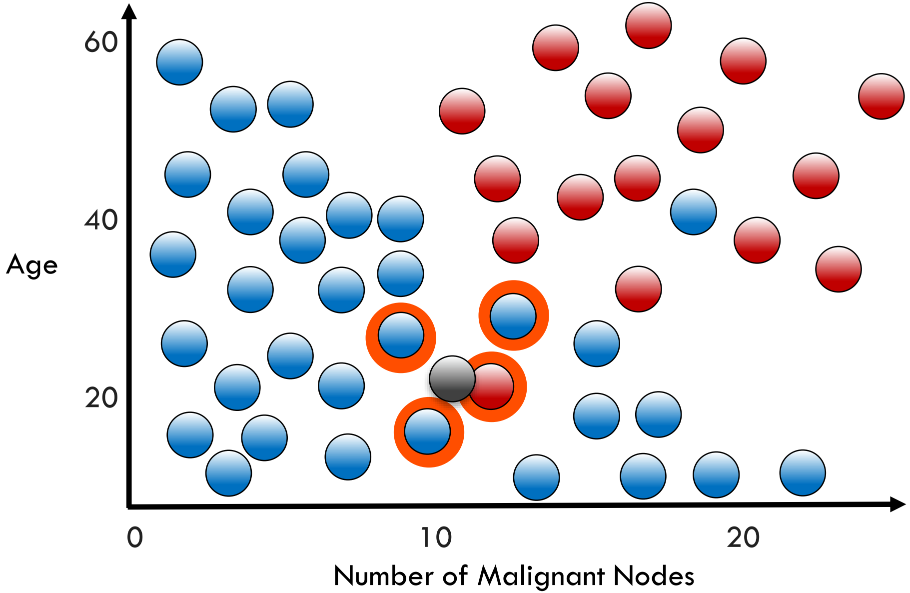
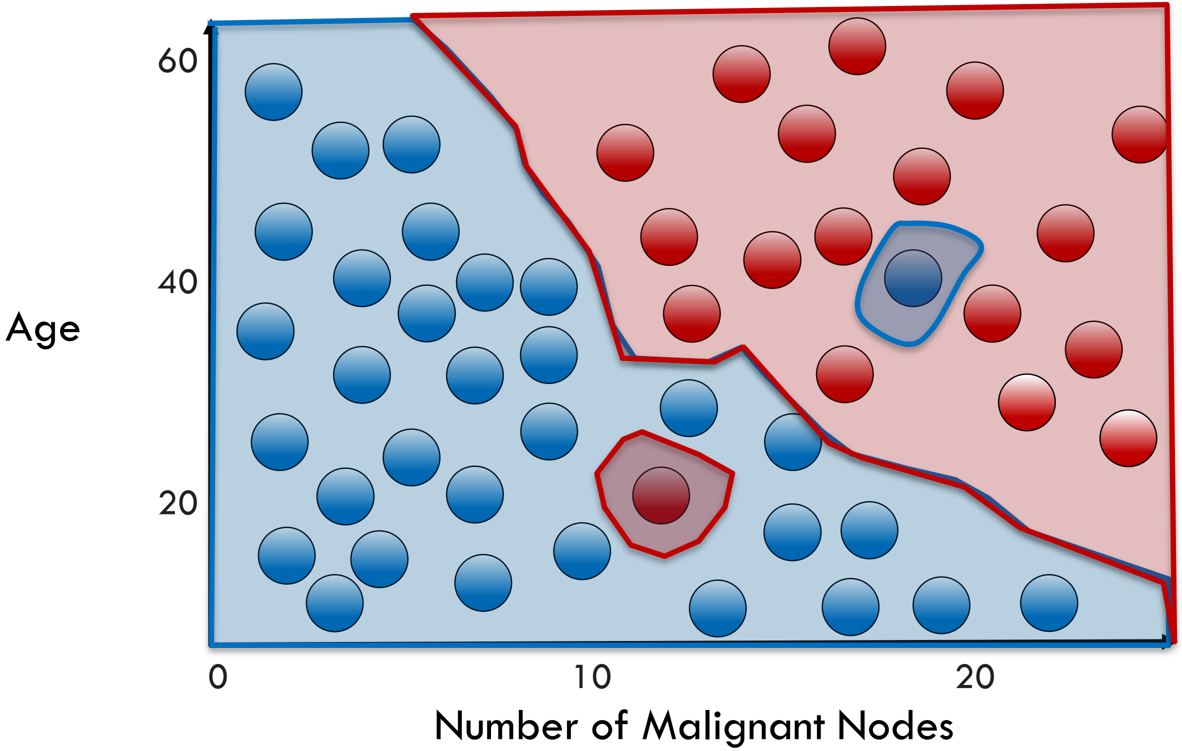
- Correct value for 'K'
- How to measure closeness of neighbors?



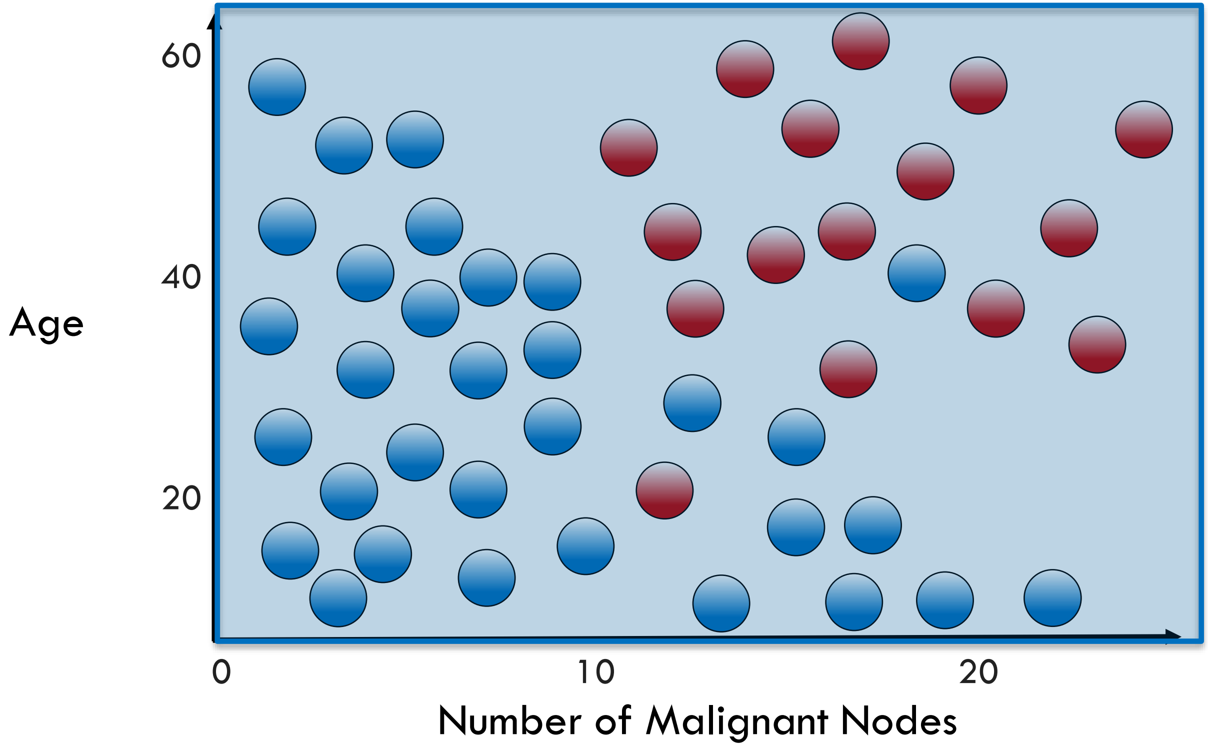
K Nearest Neighbors Decision Boundary

K = 1

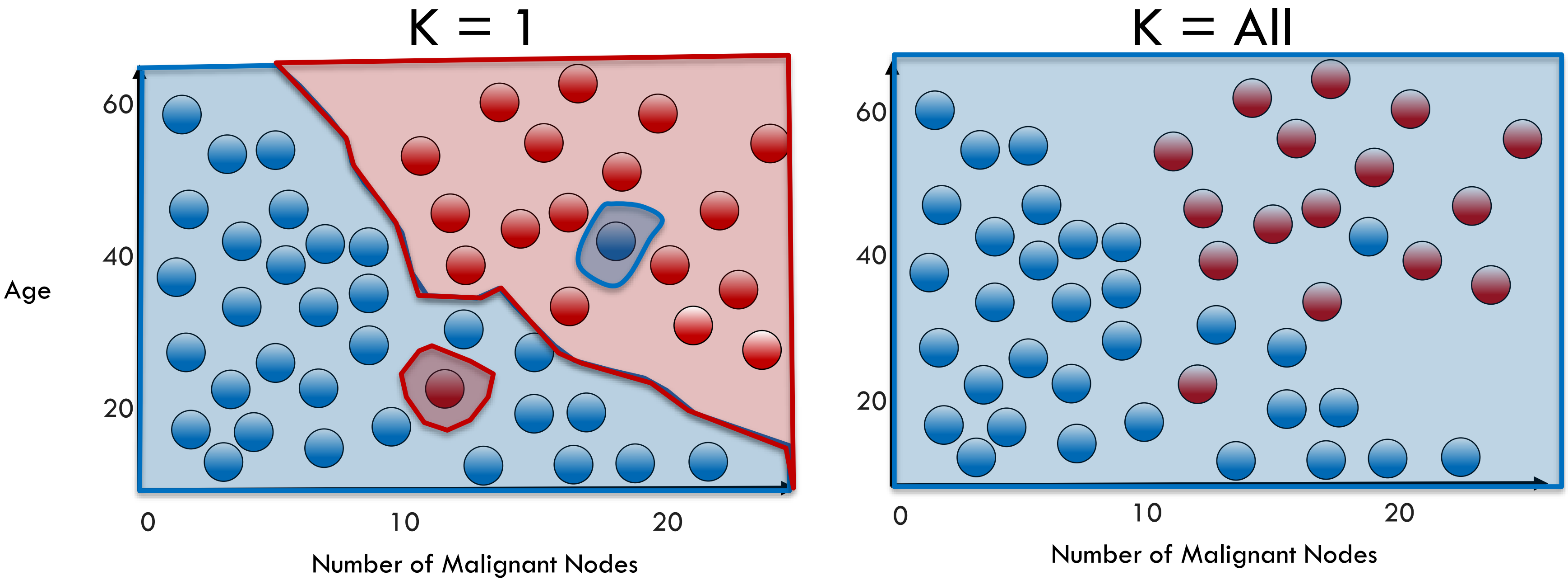


K Nearest Neighbors Decision Boundary

K = All



Value of 'K' Affects Decision Boundary



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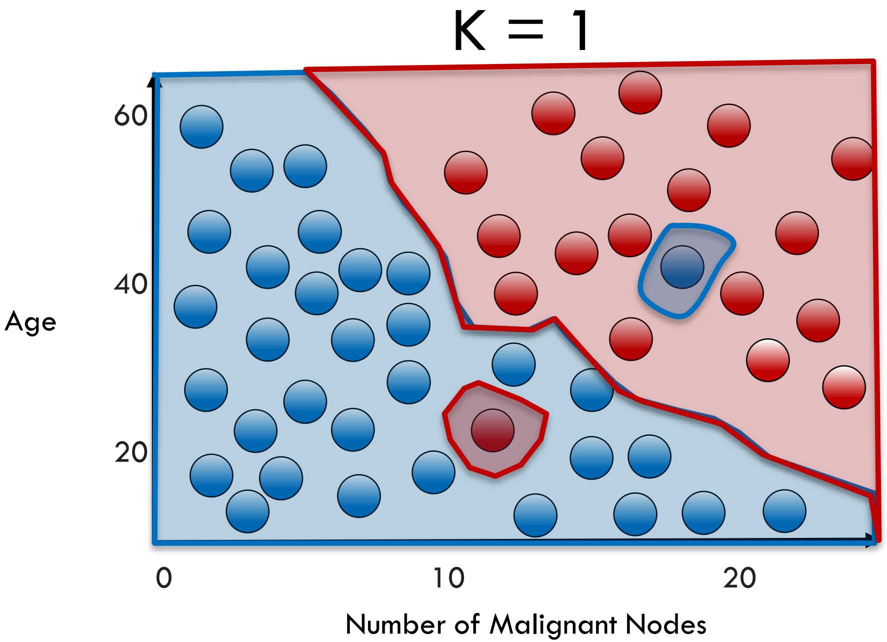
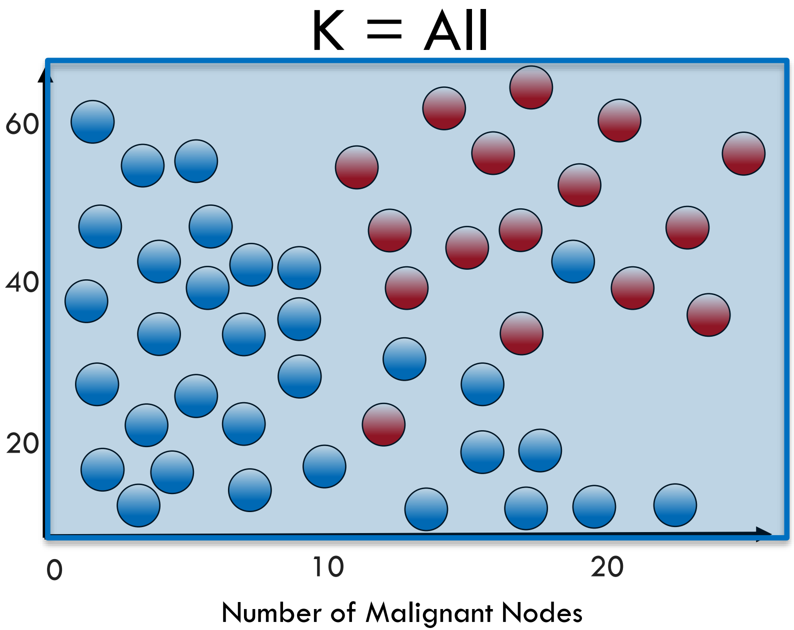
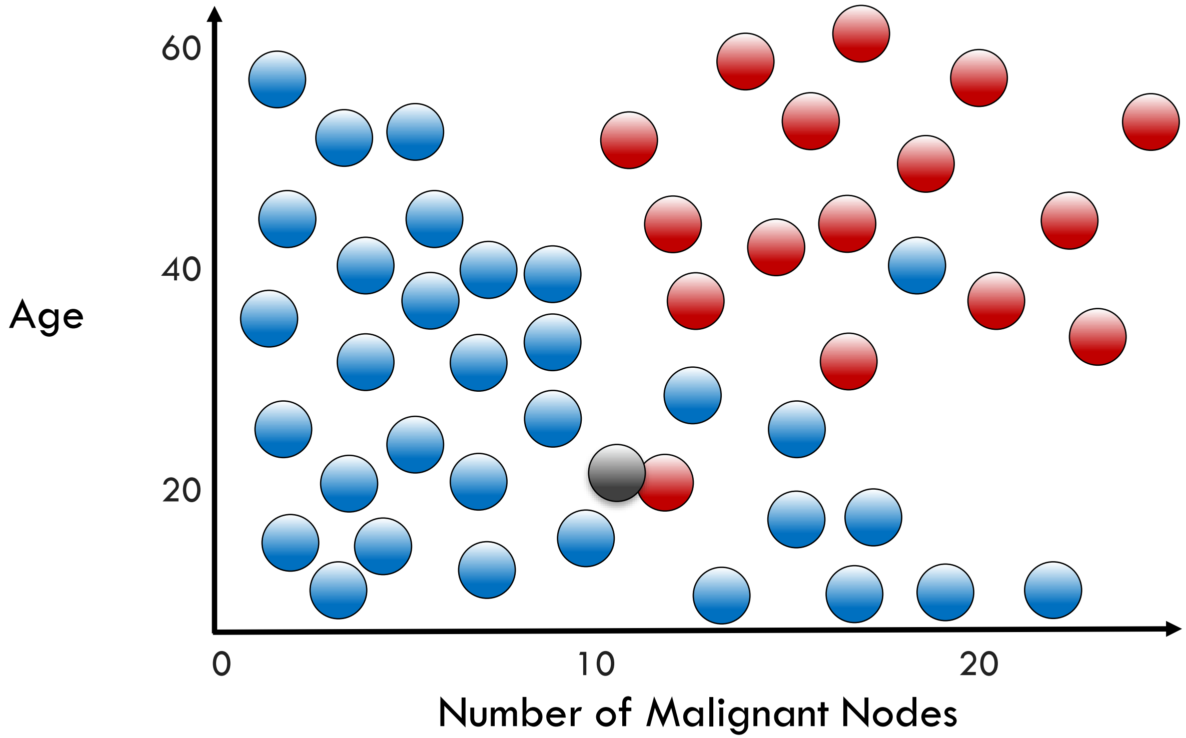




Methods for determining 'K' will be discussed in next lesson

Measurement of Distance in KNN



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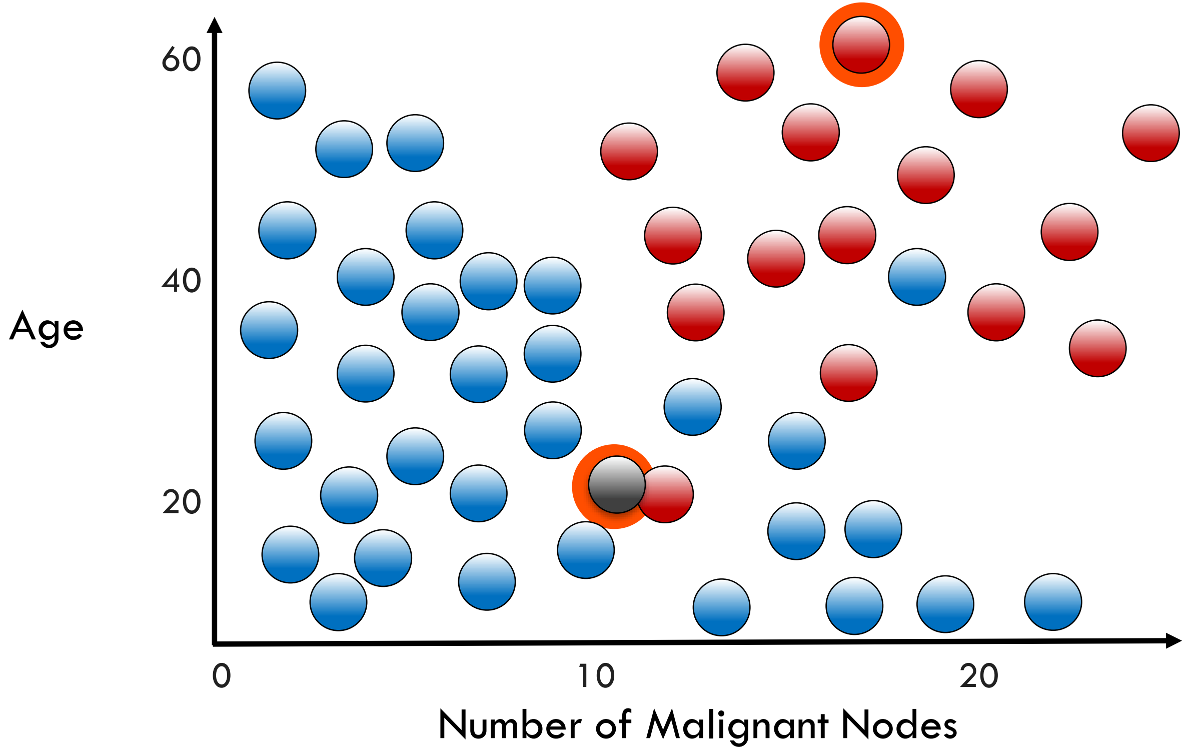
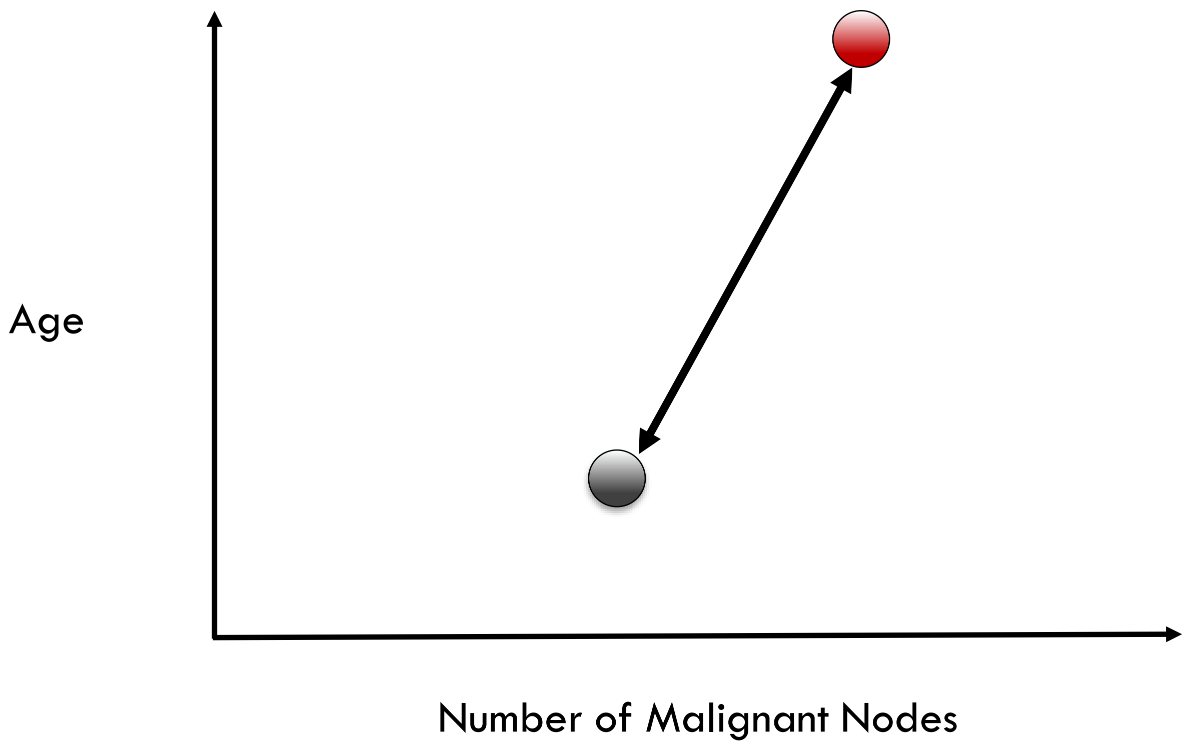
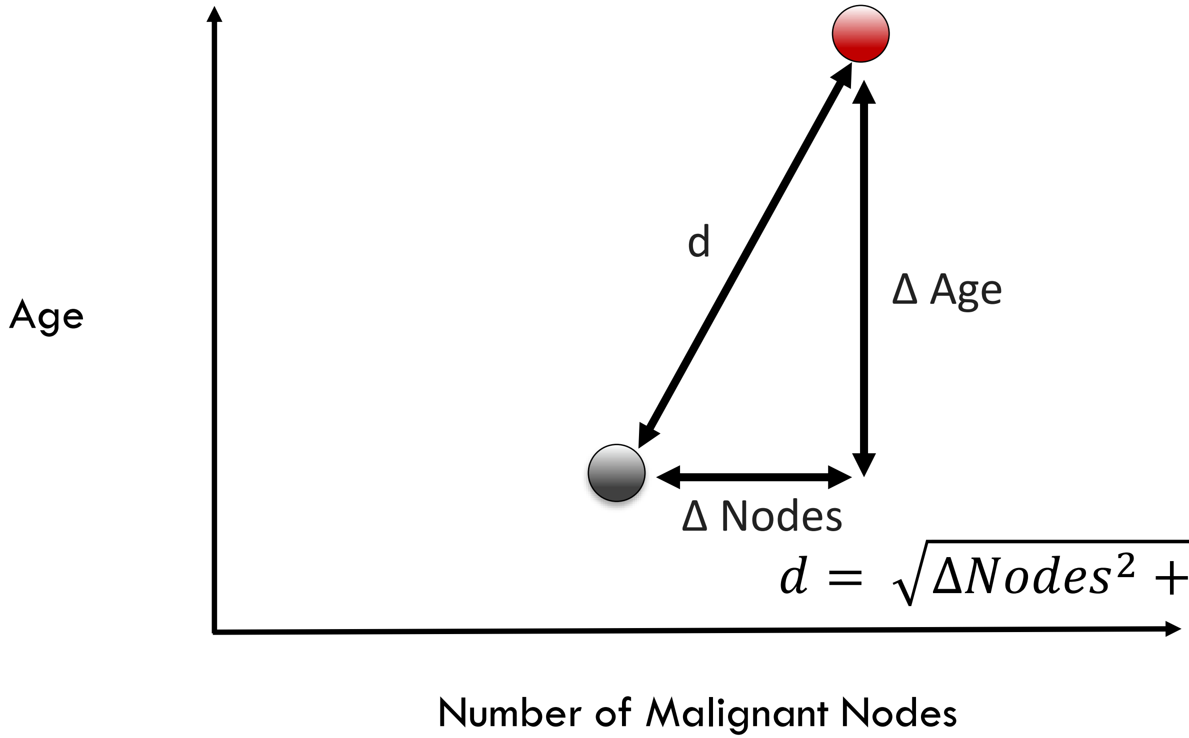


Euclidean Distance

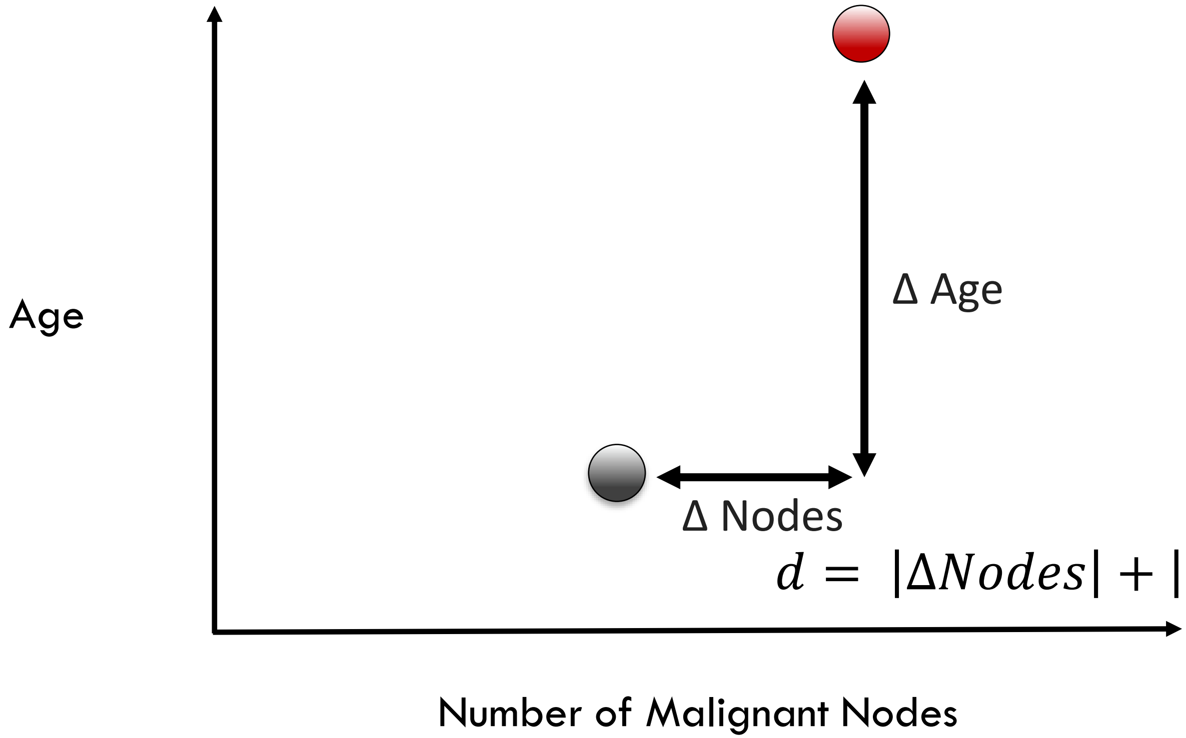


Euclidean Distance (L2 Distance)



$$d = \sqrt{\Delta Nodes^2 + \Delta Age^2}$$

Manhattan Distance (L1 or City Block Distance)



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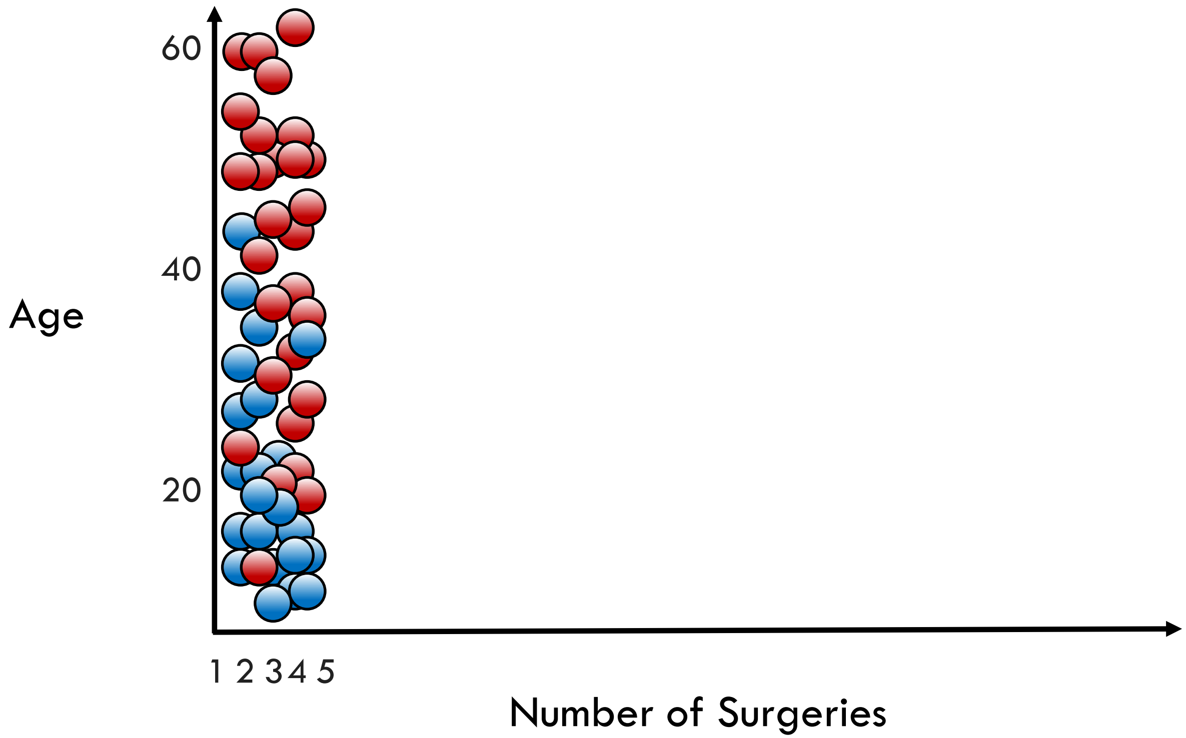
Scale is Important for Distance Measurement



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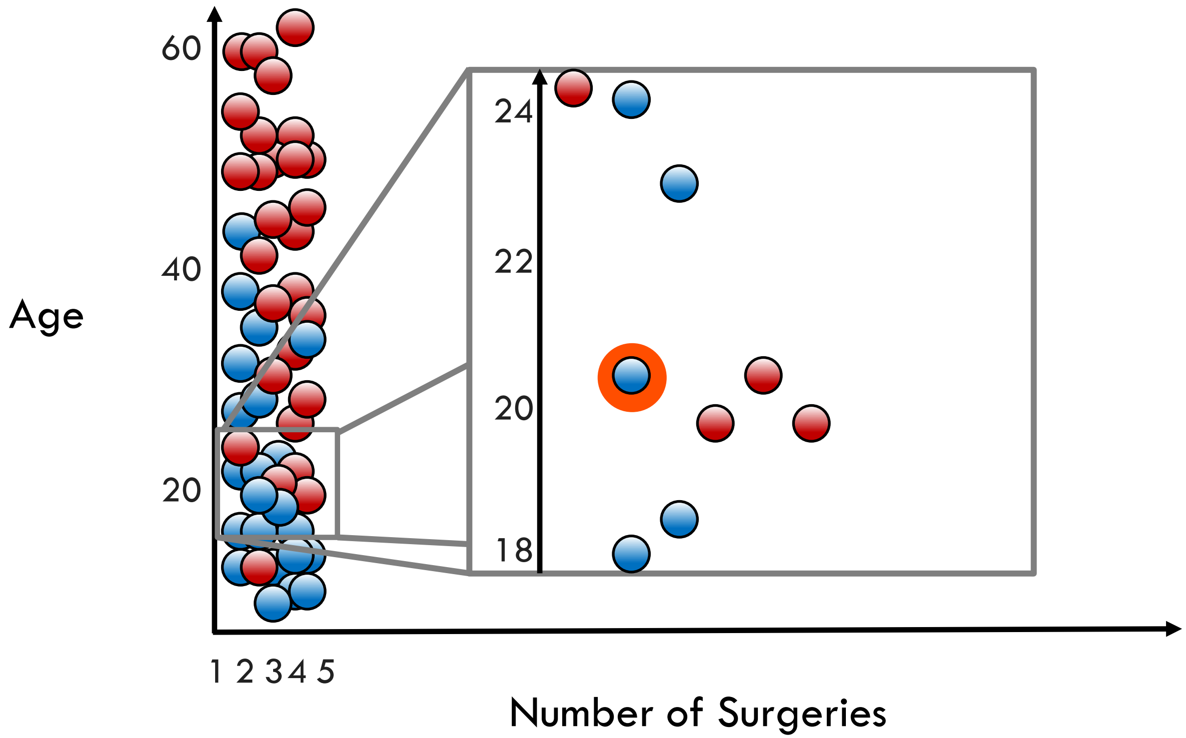

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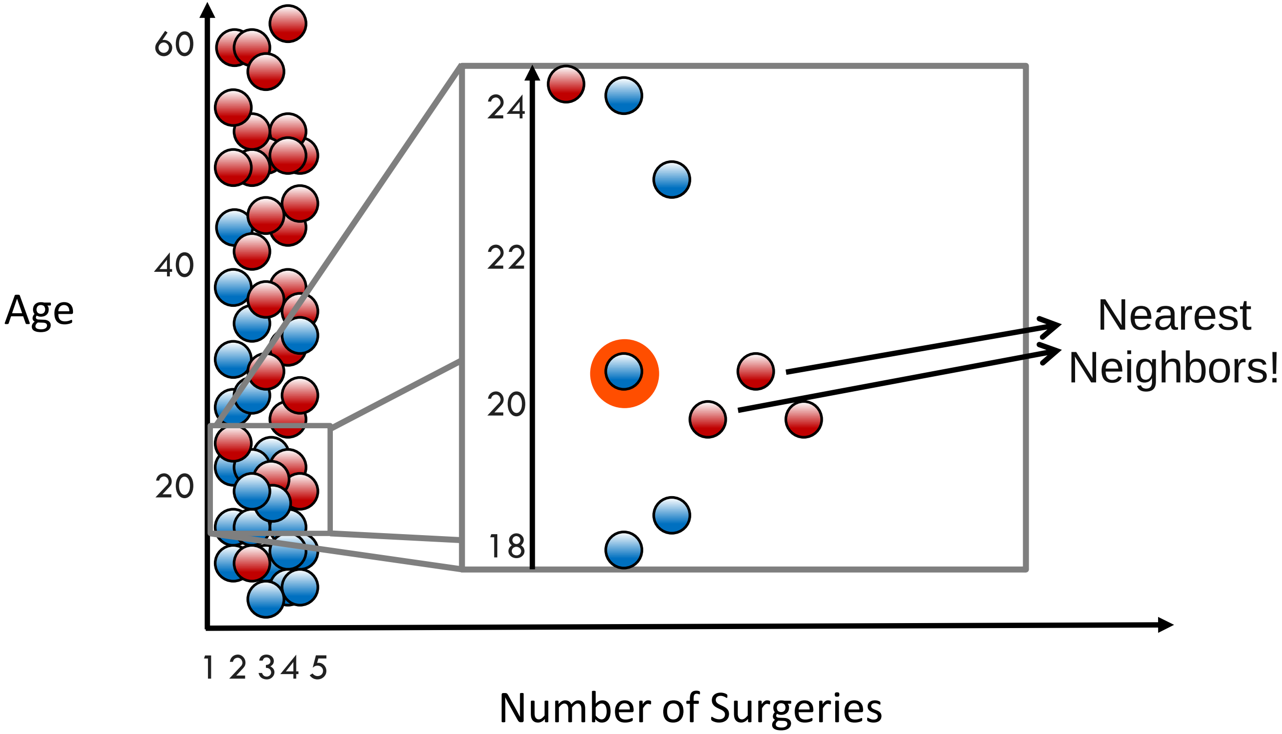
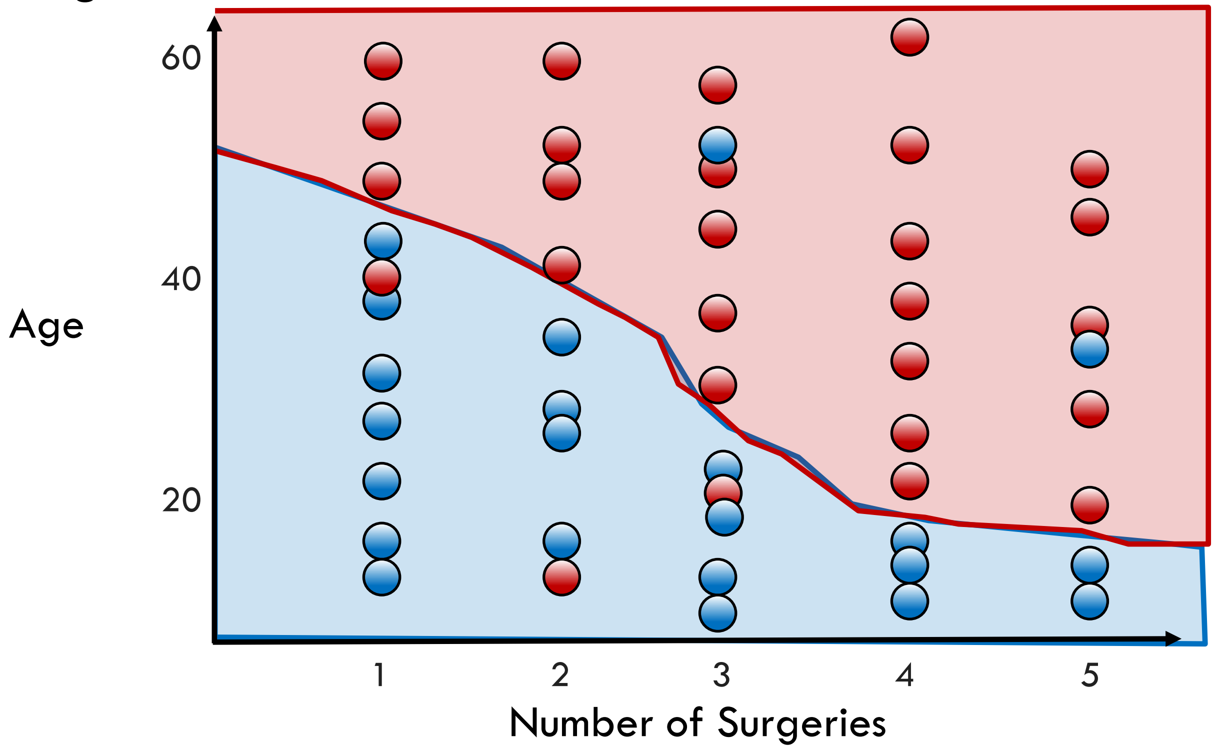


Scale is Important for Distance Measurement

"Feature Scaling"



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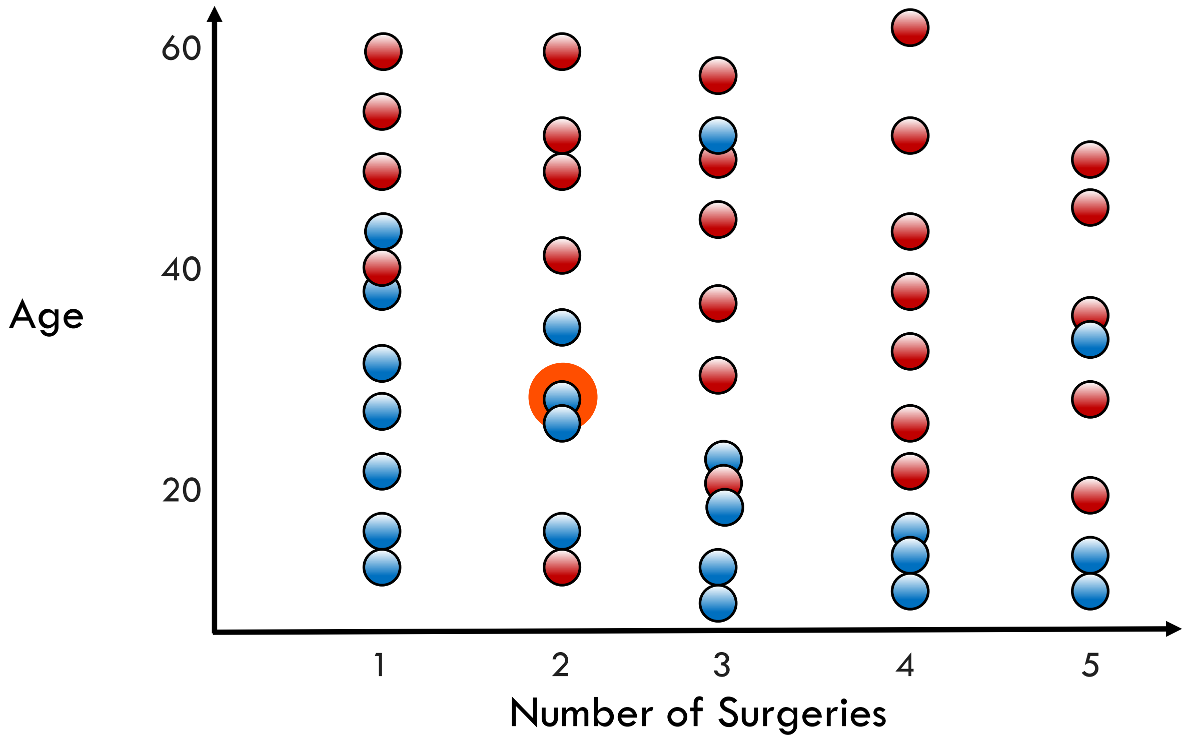

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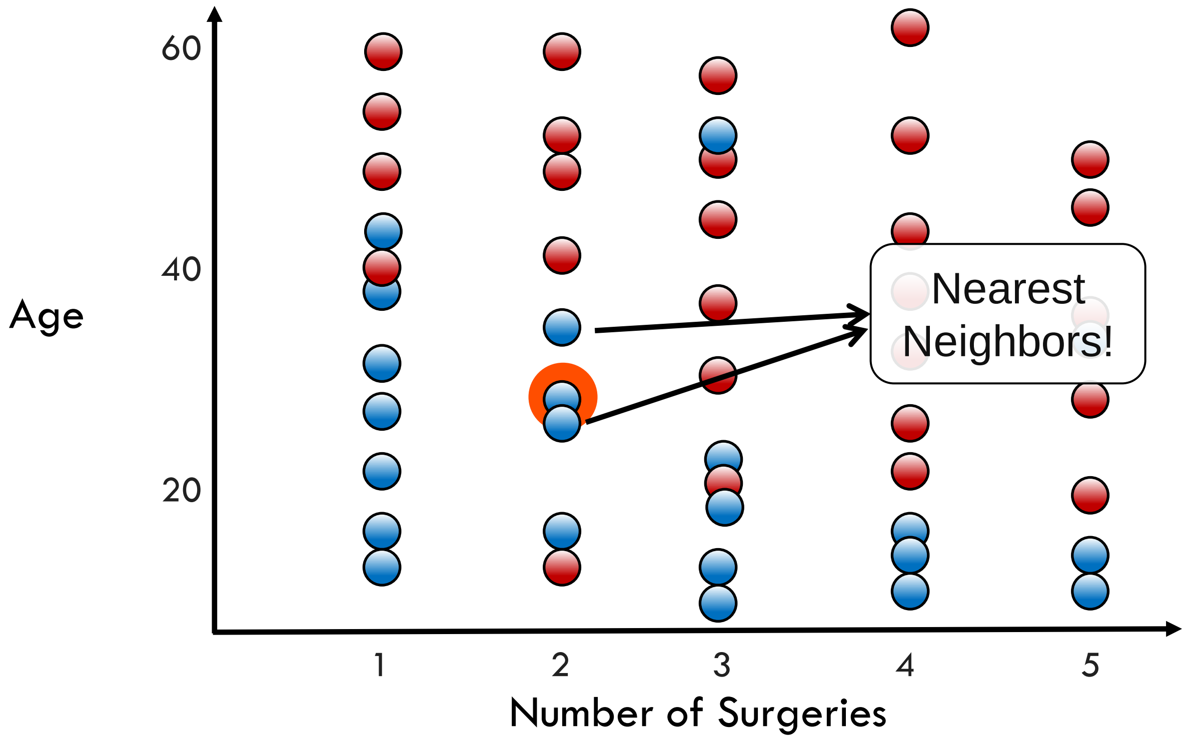

Comparison of Feature Scaling Methods

- **Standard Scaler:** mean center data and scale to unit variance
- **Minimum-Maximum Scaler:** scale data to fixed range (usually 0–1)
- **Maximum Absolute Value Scaler:** scale maximum absolute value

Feature Scaling: The Syntax

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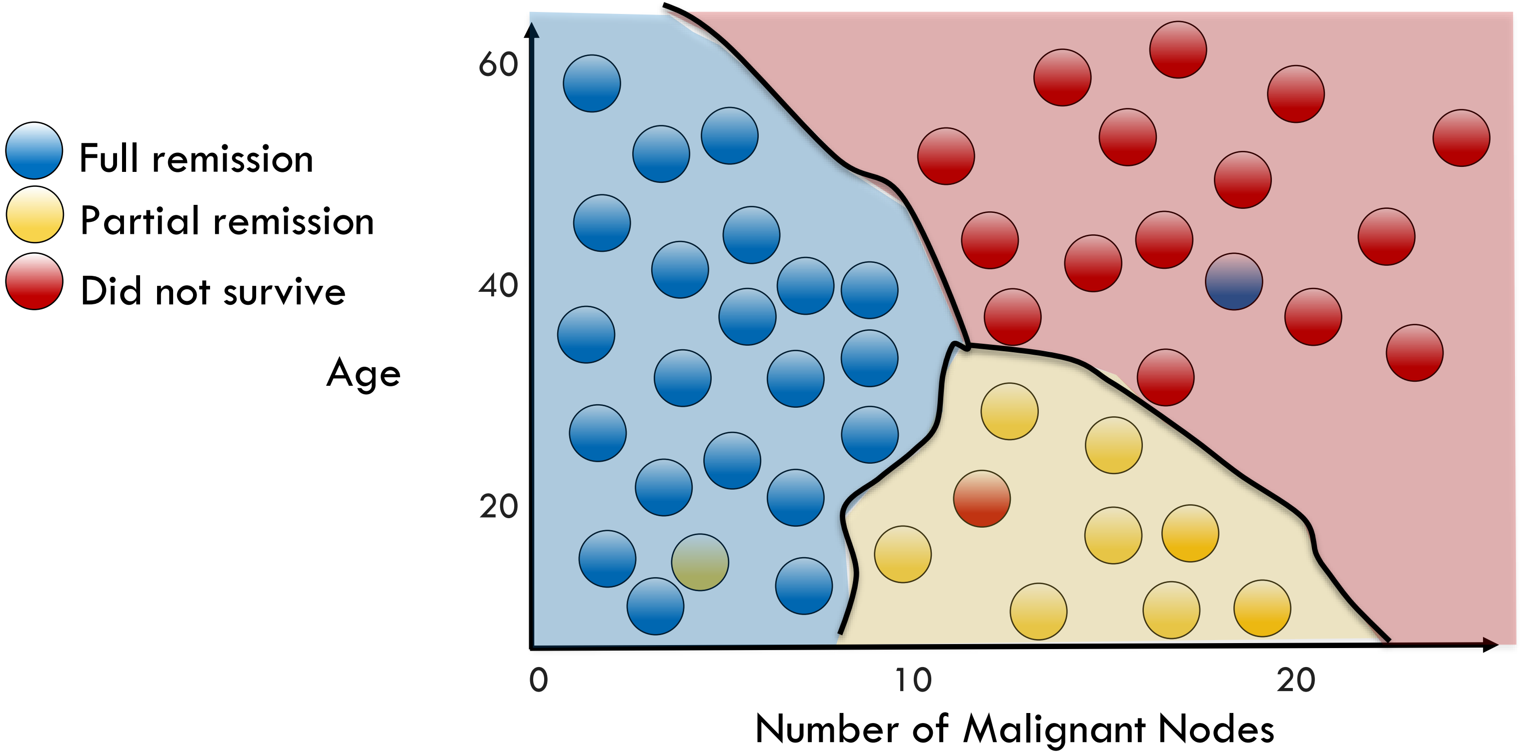
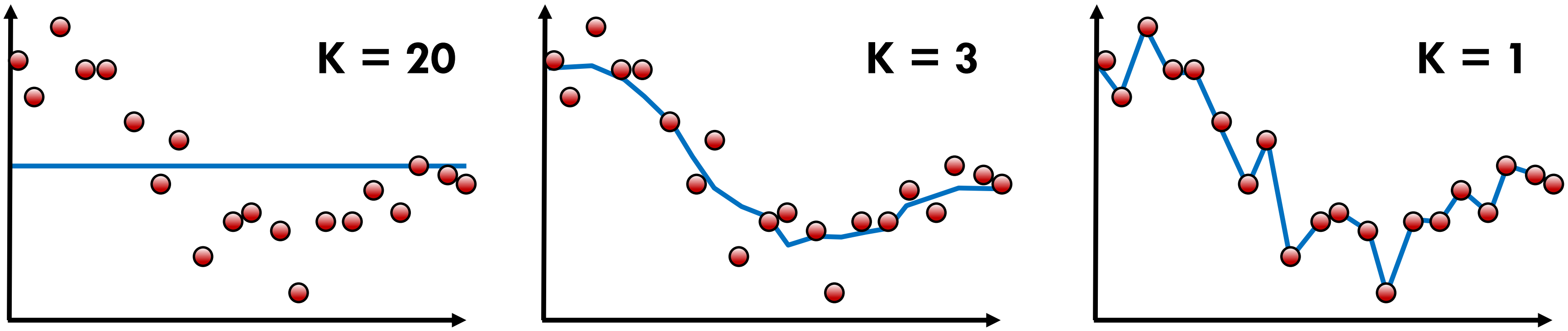
Other scaling methods exist: MinMaxScaler, MaxAbsScaler.

Multiclass KNN Decision Boundary

K = 5



Regression with KNN



Characteristics of a KNN Model

- Fast to create model because it simply stores data
- Slow to predict because many distance calculations
- Can require lots of memory if data set is large

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The fit and predict/transform syntax will show up throughout the course.

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Regression can be done with KNeighborsRegressor.